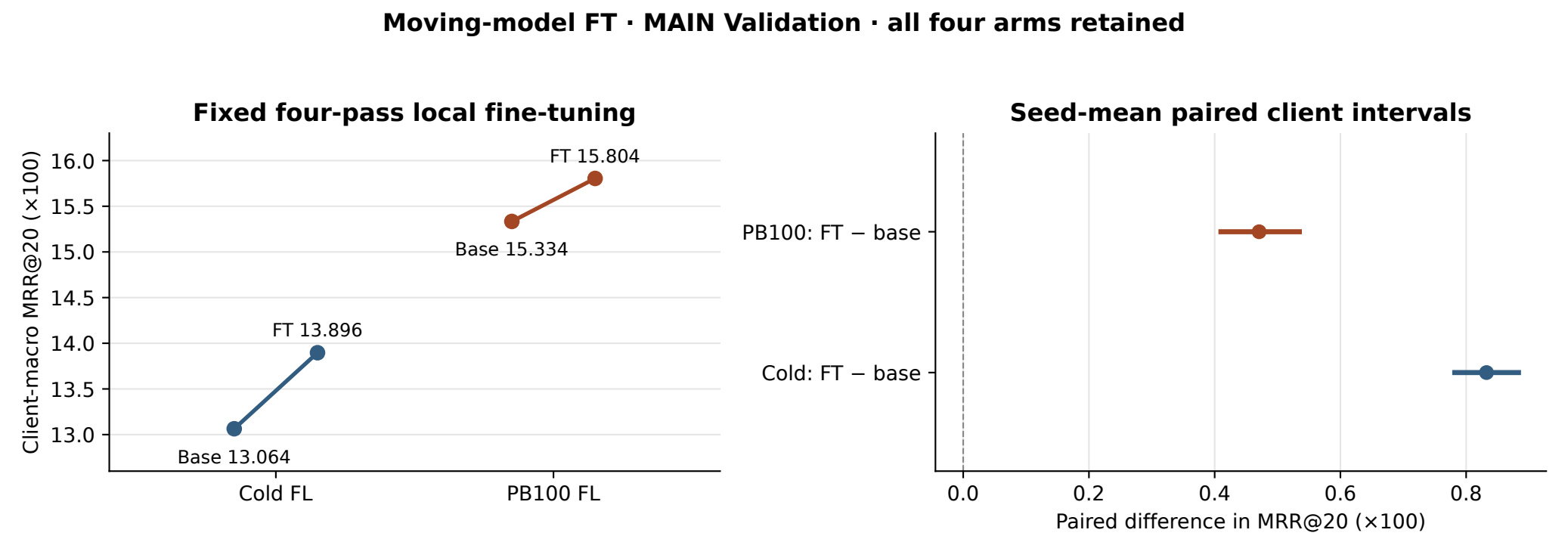
Moving-model FT · MAIN Validation · all four arms retained
Fixed four-pass local fine-tuning
Client-macro MRR@20 (×100)
16.0
15.5
15.0
14.5
14.0
13.5
13.0
FT 15.804
Base 15.334
FT 13.896
Base 13.064
Cold FL
PB100 FL
Seed-mean paired client intervals
PB100: FT − base
Cold: FT − base
0.0
0.2
0.4
0.6
0.8
Paired difference in MRR@20 (×100)
40,764 users including 46 OOV-only users · 519,714 E2E decisions · fixed training seeds 2026/2027
1,000 paired client draws · pointwise 95% intervals · post-hoc exploratory; conditional on fixed data/seeds.